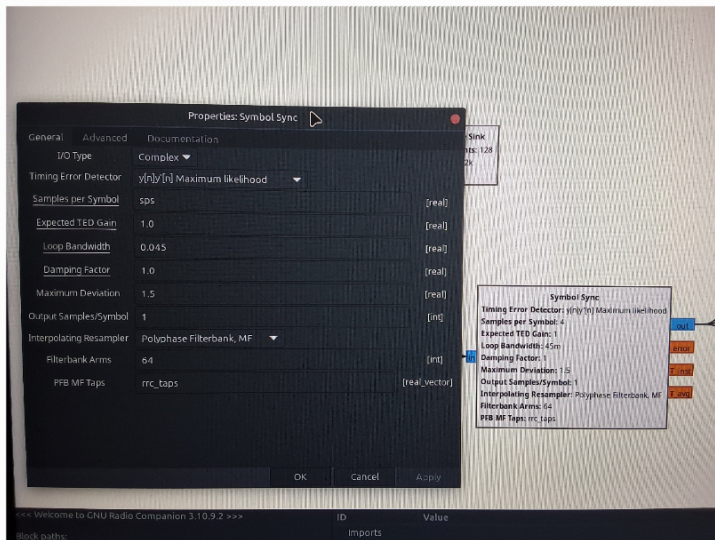
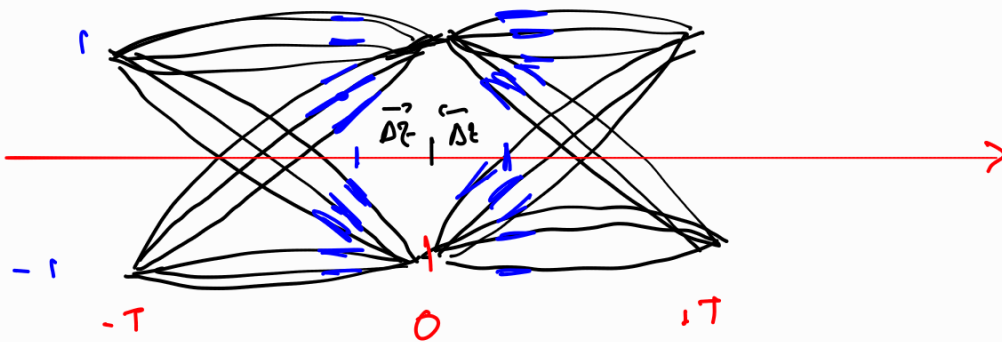


Symbol sync



- This used to find the optimal point to take the sample.
- method used: $y[n]y'[n]$ maximum likelihood eye diagram: (multiple symbols stacked)



- $y[n]$ is the signal and $y'[n]$ is slope.
- error from 0s (sample point) is given by,

$$e = y \cdot y'$$

if the point is to right

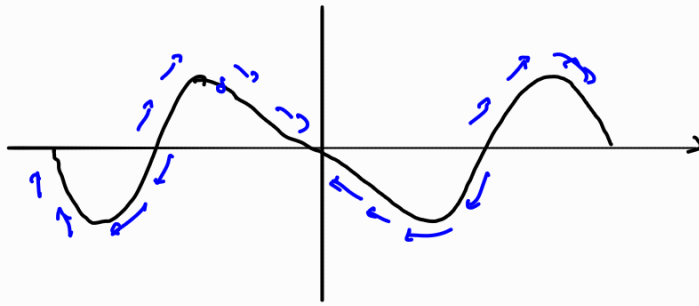
„ „ „ „ „ left

$y, y' > 0$ (refer to eye diag)
or $y, y' < 0$

$y > 0, y' < 0$
or

$y < 0, y' > 0$

so if so right $e > 0$ else $e < 0$



• calc error so converge to nearest optimal sample point?

~~polyphase filter bank does this by,
let $\dot{y} = y * d$ ← derivative system~~

$$\begin{aligned} \text{then } e &= (x * RRC) * d \\ &= x * (RRC * d) \\ &= x * \dot{RRC} \end{aligned}$$

~~polyphase calc $\dot{RRC}$ and applies it, so no need of a extra matched filter,~~

taps:

$$y[n] = b_0 x[n] + b_1 x[n-1] + \dots + b_n$$

- coefficient $b_0 - b_n$ are taps.
- instead of doing calc every time to apply a filter these are precalculated to save computation.

interpolating resampler.

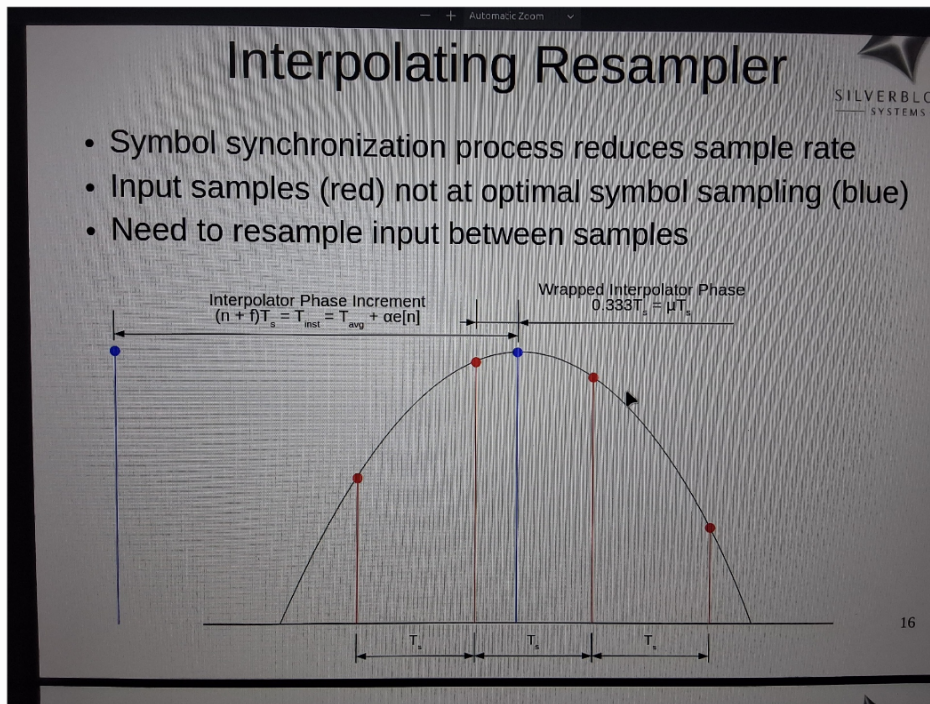
- MMSE, 8 Tap FIR

Min Mean Square Error.

- perfect sample point may happen between samples



in order find the perfect sample, software interpolates between them to find the (near) actual value. This is done by different types of interpolators.



- GR Minimum Mean Squared Error interpolators
 - Truncated, MMSE version of ideal filters
 - Bank of 129, 8 tap FIR filters
 - For μ in $\{0/128, 1/128, 2/128, \dots, 127/128, 128/128\}$
 - Each filter has a MMSE $H(\omega)$ only in $[-F_s/4, F_s/4]$
- One sided freq response and error² of the filters

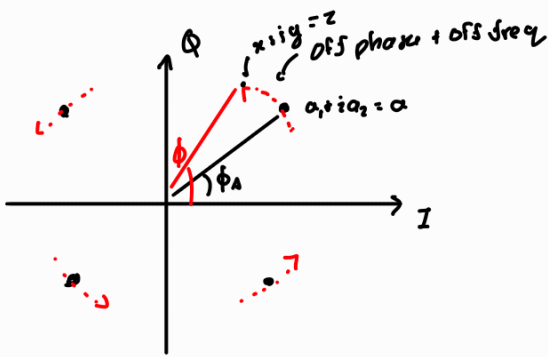
← fractional delay

- system pre calculates 8 taps of filter (n,n+7)
for $\mu = 0, \frac{1}{128}, \dots, 1$
- Timing error detector finds the required μ .
- it retrieves the matching taps and applies it for resampling.
- basically it precalculates for some delays the filter values required to get the optimal sample value.

Costas loop

- phase locked loop.

Input : $r(t) = A \cos(2\pi f t + \phi)$
 $z(t) = A e^{j(2\pi(f-f_0)t + \phi)}$ ← out of SDR



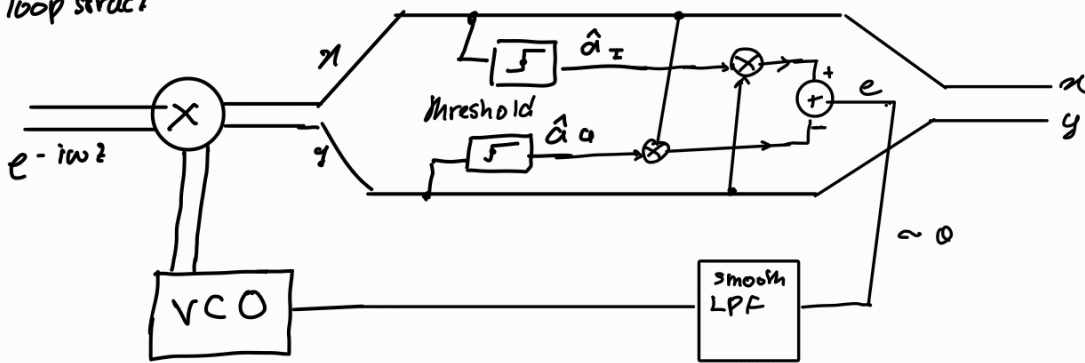
$$\theta = \phi - \phi_A$$

$$= \angle(z e^{j\phi_A}) = \tan^{-1} \left(\frac{\text{Im}\{z e^{j\phi_A}\}}{\text{Re}\{z e^{j\phi_A}\}} \right) \approx \text{Im}\{z e^{j\phi_A}\} \quad \text{if } \sin \theta \approx \theta$$

$$\approx y a_T - x a_Q$$

drive θ to zero.

Costas loop struc?



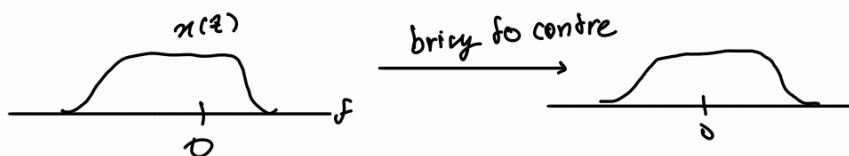
- θ drives VCO to reduce $\ln \theta$

Block

- loop bandwidth : smoothing (of the filter) BW ↑ more noise (wider)
 BW ↓ less noise (not fast enough)
 • need to balance
- recommended start : $\frac{2\pi}{100}$
- order : BPSK - 2
 QPSK - 4

FLL Band edge

FLL



make a filter : $\Rightarrow$ e is not centered
 $e > 0$ move left
 $e < 0$ " right
 $e = 0$ ✓

- coarse frequency adjustment.

(to reduce large variations so easier lock using costas)

Block:

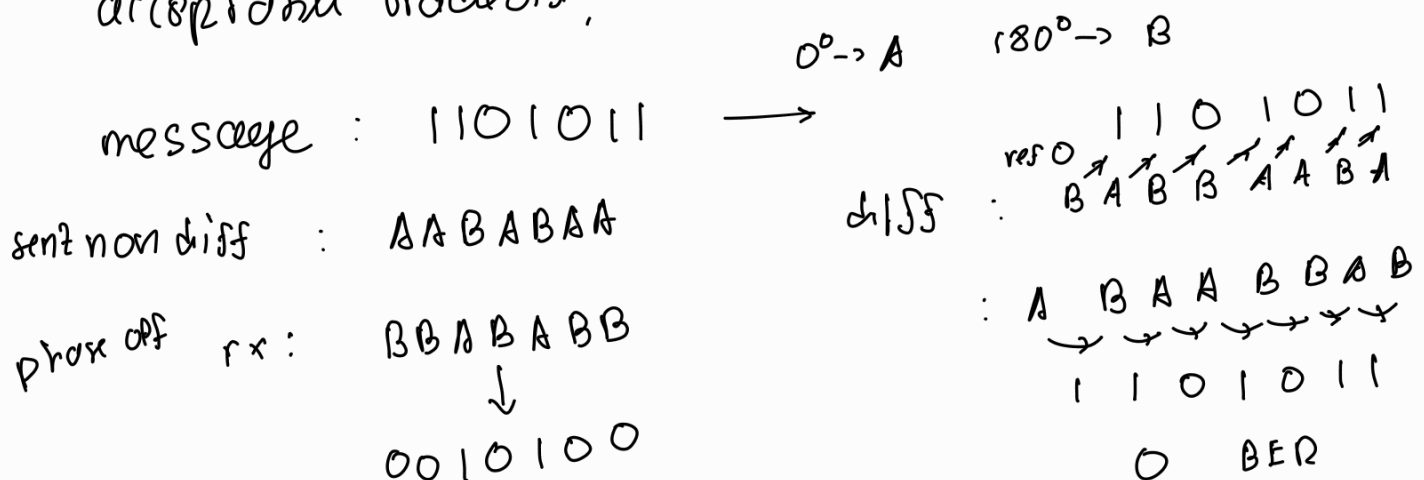
$$\text{Filter size} = \text{sps} * 2 + 1$$

$$\text{loop Bandwidth (smoothing factor for error)} \approx \frac{2\pi}{100}$$

phase ambiguity

- causes lock's to closed constellation points.
- so it can be either the true points or rotated, we cannot ensure which one supposed to go to which.
- either a unique word can be used to identify them or we can use differential encoding

we use diff encoding, but it causes the BER $\times 2$. but acceptable tradeoffs.



encode: $d_k \oplus e_{k-1}$

$\nearrow$ msg $\nwarrow$ previous encoded

decode: $r_k \oplus r_{k-1}$

$\nwarrow$ rec'd

one slip results in max 2 BER

AGC

- applied to incoming signals to have constant amplitude
- due to power fluctuations, it may drop or increase, AGC adjusts and bring it closer to required amplitude

$$\text{Gain}(n+1) = \text{Gain}(n) + \left\{ \text{Rate} \times \frac{(\text{reference} - |y(n)|)}{(\sim 1)} \right\}$$

$$\text{out} = y(n) \times \text{Gain}(n)$$

- max is set to prevent setting high gains at transition period.
- other synchronizers rely on near ideal transitions, power fluctuations may cause desynchronization. AAC ensures it outputs expected amplitudes.

Frame : preamble | sync word | Length | Payload | CRC 32 | postamble.

- Preamble : to wake up the synchronizer to decoding and sync phase.

used preamble 101010.....

- Postamble : to flush out any remaining data inside the filters, buffers.

used postamble 101010.....

- Sync word : a word to synchronize and identify the frame start. Data is autocorrelated and when it peaks filter end is considered as frame start

used sync word : CCSDS : 0x1ACFFC1D

- standard 32 bit sync word used.

- length : payload length

Rx mechanism :

- rx bits are shifted through a "shift register" and autocorrelated

- when found payload length is extracted.
- If under max payload length, CRC is checked.
- if CRC is validated then payload is extracted and outputed.